Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C083 **Surf1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	No gene in this candidate network	**F1 Surf1** GenAge 636 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16172499	**M1 Surf1** GenAge 637 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17210671
Direct pairs: F1–M1 (1:1)				
C025 **SWD1** 2 genes / 2 species 1 direct pair	**Y1 SWD1** GenAge 1418 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 F21H12.1** GenAge 238 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C109 **SWD3** 2 genes / 2 species 1 direct pair	**Y1 SWD3** GenAge 1586 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 wdr-5.1** GenAge 1000 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 20555324, 22012258	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:1)				
C118 **TAL1** 2 genes / 2 species 1 direct pair	**Y1 TAL1** GenAge 1449 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19030232	**W1 Y24D9A.8** **tald-1** GenAge 1939, 2213 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24662282, 28355222	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.